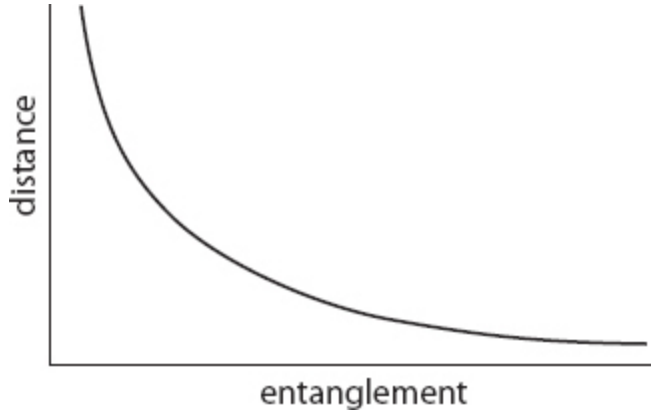
We can describe this construction in equivalent but slightly less formal terms. Pick out two of our spacetime degrees of freedom. They will generally have some entanglement between them. If they were modes of vibrating quantum fields in the vacuum state, we know exactly what that degree of entanglement would be: it would be high if they were nearby, and low if they were far away. Now we are simply thinking the other way around. If the degrees of freedom are highly entangled, we *define* them to be nearby, and the farther and farther away, the less entangled they are. A metric on space has emerged from the entanglement structure of the quantum state.

Thinking this way is a bit unusual, even for physicists, because we're used to thinking of particles moving through space, while taking space itself for granted. As we know from the EPR thought experiment, two particles can be completely entangled no matter how far away they are; there's no necessary relationship between entanglement and distance. Here, however, we're not talking about particles but about the fundamental building-block degrees of freedom that make up space itself. Those aren't entangled in any old way; they are strung together in a very specific structure.*



Now we can use Jacobson's trick with entropy and area. Knowing the area of every surface in our network gives us a geometry, and knowing the entropy of each region tells us something about the energy in that region. I've been involved with this approach myself, in papers from 2016 and 2018 with my collaborators ChunJun (Charles) Cao and Spyridon Michalakis. Closely related ideas have been investigated by Tom Banks, Willy Fischler, Steve Giddings, and other physicists who are willing to contemplate the idea that spacetime isn't fundamental, but emerges from the wave function.